

IMO 2026 — Autonomous Model Comparison

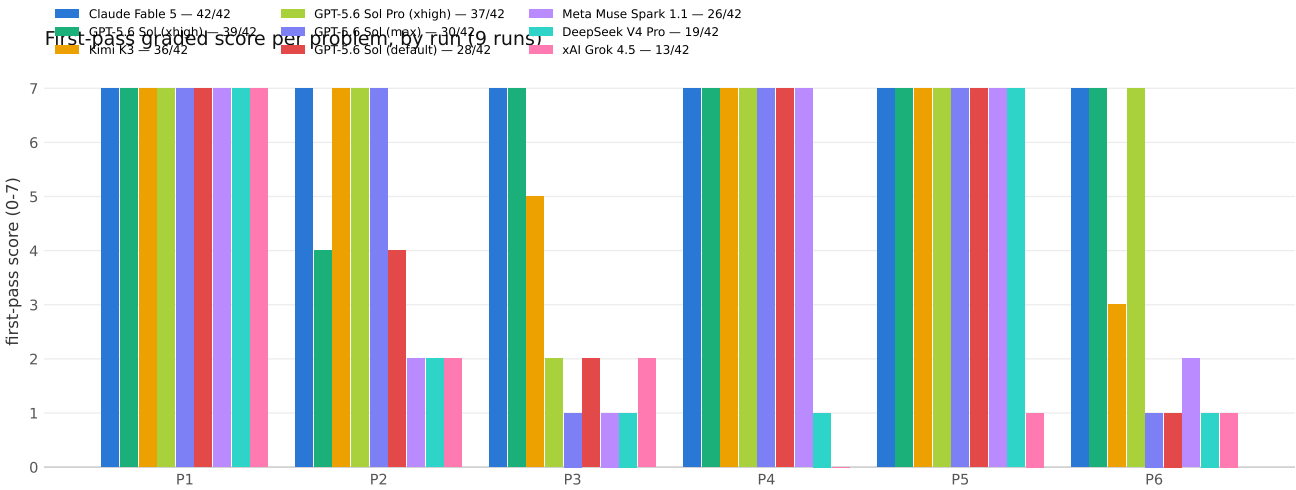
Independently graded results across nine runs (seven models)

One identical minimal agent harness for every run · IMO 0–7 grading by independent verifier agents, claims checked symbolically and by machine

Final graded scores

Run	P1	P2	P3	P4	P5	P6	Total
Claude Fable 5 (default high)	7	7	7	7	7	7	42/42
GPT-5.6 Sol (xhigh) [†]	7	7	7	7	7	7	42/42
Kimi K3 (default) [†]	7	7	7	7	7	7	42/42
GPT-5.6 Sol Pro (xhigh) [§]	7	7	2	7	7	7	37/42
GPT-5.6 Sol (max)	7	7	1	7	7	1	30/42
GPT-5.6 Sol (default)	7	4	2	7	7	1	28/42
Meta Muse Spark 1.1 (xhigh)	7	2	1	7	7	2	26/42
DeepSeek V4 Pro (max) [‡]	7	2	1	1	7	1	19/42
xAI Grok 4.5 (xhigh) [‡]	7	2	2	0	1	1	13/42

[†] reached via repair rounds seeded with reviewer defect findings (no external solutions). [§] GPT-5.6 Sol Pro and GPT-5.6 Sol (**max**) ran single-pass, no reviewer feedback; Sol Pro’s P6 got one fair fresh retry after a degenerate 2-turn null first session (its P2 was also interrupted twice by an orchestration/credit failure and re-run to completion). [‡] Grok and DeepSeek needed a harness guard to force them to actually call the write-tool. Claude Fable 5 is the only run graded 42/42 on its first pass.



The two controlled follow-up questions

- **Does effort beyond **xhigh** help?** No. The same base `gpt-5.6-sol` at `max` effort (the highest OpenRouter exposes) scored **30/42** single-pass — *below* its own **39/42** first-pass at **xhigh**. More reasoning budget reshuffled which problems fell (`max` uniquely cracked the P2 geometry that **xhigh** first missed, but then dropped P6 and P3) rather than raising the total.

- **Does the higher gpt-5.6-sol-pro tier help at equal effort?** No. At the same `xhigh` effort, Sol Pro scored **37/42** — also below base Sol’s **39/42** first-pass, sinking the P2 geometry and the P3 combinatorics capstone.
- **Net:** on these single runs both “upgrades” landed in the low-30s where base-`xhigh` reached 39 — i.e. within run-to-run variance, not a monotone climb. Neither knob reliably closed the two hardest problems (P2, P3).
- **Right answer, wrong (or no) proof.** Sol Pro determined P3’s *correct* answer $c = 2^n / (2^{n+1} - 1)$ (verified by exact minimax) yet scored only 2/7 for lack of a proof — still categorically better than Muse Spark 1.1 and DeepSeek, which confidently proved the *false* $c = (n + 1) / (2n + 1)$.

Time and cost (all rounds included)

Run	Final	Total time	Cost (USD)	Output tokens
Claude Fable 5	42/42	2.5 h	\$51.05	705k
GPT-5.6 Sol (<code>xhigh</code>)	42/42	3.8 h	~\$20.54	236k
Kimi K3	42/42	17.4 h	~\$31.40	1,545k
GPT-5.6 Sol Pro (<code>xhigh</code>)	37/42	2.5 h	~\$16.02	367k
GPT-5.6 Sol (<code>max</code>)	30/42	2.1 h	~\$23.43	204k
GPT-5.6 Sol (default)	28/42	1.0 h	~\$4.04	80k
Meta Muse Spark 1.1	26/42	2.3 h	~\$4.45	761k
DeepSeek V4 Pro	19/42	8.9 h	~\$6.05	2,013k
xAI Grok 4.5	13/42	3.2 h	~\$14.60	606k

Anthropic API metering for Claude Fable 5; OpenRouter list pricing (Sol / Sol Pro: \$5/M prompt, \$30/M completion) for the GPT-5.6 Sol runs, Muse, Grok, DeepSeek; Kimi K3 estimated from logged tokens. Sol (`max`) cost more than Sol Pro despite fewer output tokens because its P2 alone churned 3.3M prompt tokens over the full 80-turn budget.

Caveats

- Graders are Claude-based agents, not human medalists.
- All nine runs share the identical harness. The four GPT-5.6 Sol runs isolate two clean variables: effort on the base model (`medium`→`xhigh`→`max`) and tier at fixed `xhigh` effort (base vs `-pro`). Single runs per setting — treat low-30s-vs-39 gaps as noise-band, not a precise ranking.
- Full per-turn audit trails and per-problem grader verdicts accompany this document; see REPORT.md for attempts/time/tokens/cost.